

WEEKLY PROBLEM

EXERCISE.

Quantum Ring Annihilation

Particles are arranged in a circular ring. Each particle is either:

- Positive (P)
- Negative (N)

At each step (simultaneously):

1. Every P with an N immediately to its right annihilates.
2. Every N with a P immediately to its left also annihilates.
3. Marked particles are removed.
4. The ring closes and repeats.

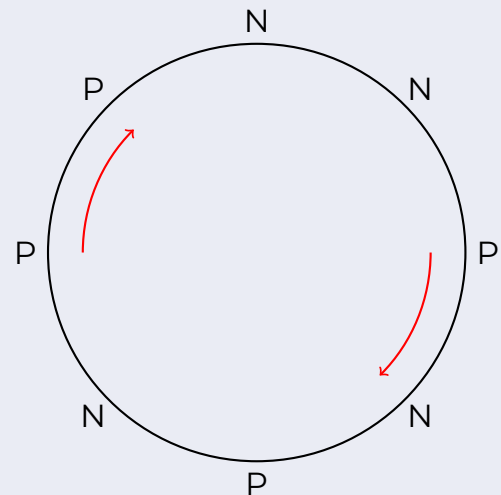


Fig: Annihilation direction ($P \rightarrow N$)

You may insert exactly one particle (P or N) anywhere. Goal: be the **unique last survivor**.

(A) Hidden Stability Zones

P, N, N, P, P, N, P, N

Find all insertion positions and choices such that:

- You are never annihilated in any round
- You are the only particle remaining at the end

(B) Adversarial Insertion

P, N, P, N, P, N

You insert first by choosing position and type, opponent inserts next anywhere.

Can you guarantee being the **unique survivor** regardless of opponent's move?

(C) Function-Based Classification

Particles labeled from 1 to 100.

- P if $\lfloor \sqrt{n} \rfloor$ is even
- N otherwise

Find all (k, type) such that inserting after position k ensures you are the **last survivor**.



(Scan this to submit solution!)

Top Submissions for last Week's Problem (WP2604):

1. **Swastik Mishra** - CS25B1039